Present-Day Surface TiO_2 and Lunar Crustal Magnetic Anomalies

An underpowered test of spatial co-location, not a temporal test of an intermittent dynamo

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An independent, LLM-assisted learning project — not peer-reviewed. See §8 for the assistance and responsibility disclosure.

ABSTRACT

We compare a present-day LROC WAC surface-TiO_2 map with a thresholded map of lunar crustal magnetic anomalies using grouped spatial cross-validation. The full model has PR-AUC 0.089, below an approximate six-basin antipode benchmark (0.1106) and near a matched full-model spatial-rotation null (0.1127 empirical p = 0.56); removing the TiO_2 family does not reduce performance (−0.0030). In contrast, H2 fragilely exceeds its own dedicated null (mean 0.0320, 95th 0.1068; p = 0.040 at 1/101 resolution, margin 0.0037). The earlier claim that H2 sat at null was a mismatched-null error. The failed H1 criteria are not a calibrated negative: the fitted target range is about 3,752 km versus 910 km blocks and the approximate effective sample size is 1. The amended status is `INCONCLUSIVE_LOW_POWER`. The experiment tests present-day map co-location only; it neither confirms nor refutes the temporal/thermal intermittent-dynamo mechanism of Nichols et al. (2026). A post-hoc USGS mare-domain sensitivity is positive in the mean but not statistically significant and fold-unstable. A completed direct H2 injection curve recovers only large encoded effects and leaves `adequate_power` false. A companion H1 (TiO_2-driver) injection curve, with the strength grid extended into the realistic score regime, calibrates the sensitivity of the hypothesis actually under test: at an illustrative strength-1.0 anchor its power is 0.400 (Wilson 95% CI 0.246–0.577), so the low-power classification is demonstrated, not asserted.

WHAT THIS DOES AND DOES NOT SHOW

Mechanism (untouched). Nichols et al. propose a *temporal* mechanism — sinking Ti-rich cumulates driving short dynamo episodes. With no magnetization ages or paleointensities, this study cannot confirm or refute it; the mechanism may be entirely correct. **Mappability (tested).** A separate, narrower question — whether present-day orbital surface TiO_2 is a usable *global map proxy* for anomaly location at 30 km — for which this pipeline did not detect incremental predictive value. **Not a disproof.** The injection test recovers even a strong planted TiO_2 signal only 40% of the time (target 80%) with effective regions ≈1, so failure to detect is not evidence of absence — hence `INCONCLUSIVE_LOW_POWER`, not "decoupled". Scale/ecological inference, impact demagnetization, and downward-continuation resolution blur could each hide a real coupling in these data.

1 Scientific scope

Nichols et al. (2026) propose that deep Ti-rich cumulate melting altered core heat flow and produced short dynamo episodes associated with high-Ti volcanism. That is a claim about **timing**. A global dynamo can magnetize any rock cooling during an active interval.

This repository has no magnetization ages, eruption-age measurements, source-depth composition, or paleointensity chronology. It compares two modern maps: optical surface regolith TiO_2 and surviving crustal field strength. The historical code label "H1" therefore denotes a **candidate spatial proxy inspired by** Nichols, not the Nichols mechanism itself.

Candidate surface proxy	H2 benchmark

Present-day TiO_2 and 25/50/100 km neighborhood features. A map association would be descriptive and would not identify dynamo timing.

Distance to antipodes derived from six approximate basin centers/radii. This is a literature-motivated benchmark, not exact ground truth. It is recovered against its matched H2-only rotation null.

2 Data and measurement limits

Five 1° grids (about 30 km horizontal resolution per degree at the equator) are aligned in a lunar equal-area projection: Kaguya/Lunar Prospector surface magnetic field, LROC WAC TiO_2, GRAIL gravity and crustal thickness, and USGS geologic units. Provenance manifests and SHA-256 hashes identify the bytes used.

The TiO_2 retrieval is a present-day optical-regolith product calibrated for mare composition. The primary footprint checks latitude, nodata, numeric range, and cross-layer coverage but is not a mare/highlands validity mask. Values in highlands are out-of-domain for this inference unless independently validated. A post-hoc USGS terrain sensitivity is reported below. The magnetic source can be older and deeper than the observed surface.

The primary target thresholds $|B|$ at 5 nT. It has about 286 positives among 20,556 pixels (1.4%), clustered in a few provinces. Thresholding discards the continuous field magnitude; pixel count is not independent sample size.

3 Implemented analysis

Thirteen features enter logistic-regression and XGBoost baselines. The legacy repository-plan estimate uses five-fold `GroupKFold` over 30° × 30° block labels, nested inner grouped folds for tuning, PR-AUC, a longitudinal-rotation null, a phase-randomized null, TiO_2 ablation, SHAP, a block bootstrap, and a block-size sweep.

These are reproducible computations, but the fold construction does not establish independent regions when the fitted range exceeds the block scale. SHAP describes the fitted model; it is not evidence of a physical mechanism.

3.1 Continuous-field descriptive check

A standardized ridge comparison retains `log1p(|B|)` rather than thresholding it. Under the same five blocked folds, controls-only R^2 is 0.2590 ± 0.1377 and controls+TiO_2 is 0.2712 ± 0.1366. The fold-mean incremental R^2 is +0.0121 ± 0.0052, with increments +0.0157, +0.0192, +0.0042, +0.0088, and +0.0127. This consistency is descriptive, not independent replication: the chosen folds remain inside the fitted dependence range and no row-level p-value is reported. The small increment does not establish the surface proxy or the temporal mechanism.

4 Observed metrics

0.089	0.1127	$p = 0.56$
full model PR-AUC	rotation-null mean	descriptive null test

Diagnostic	Value	Reading
Full XGBoost	0.089	fold SD ≈0.080
H2 benchmark	0.1106	matched $p = 0.040$
Full-model rotation-null mean	0.1127	$p = 0.56$
H2-only null mean / 95th	0.0320 / 0.1068	100 fold-matched rotations
TiO_2 ablation drop	−0.0030	Wilcoxon $p = 0.781$

Diagnostic	Value	Reading
Fitted range / block scale	3752 / 910 km	block does not exceed range
Approximate effective sample size	≈1	underpowered

Both rotation nulls use the same 100 unique nonidentity shifts and the same five evaluable folds as their observed statistics. Of 125 candidate shifts, 12 with only four evaluable folds were rejected; the first 100 eligible shifts were retained (13 eligible shifts were evaluated but unused). Accepted rotated prevalence ranges from 0.00769 to 0.02277 where the shifted target intersects the fixed Imbrian mask. The full and H2 feature sets are calibrated separately.

The H2 recovery is **fragile**: with 100 rotations the empirical-p resolution is $1/101 \approx 0.0099$, and the observed score clears the null 95th percentile by only 0.0037. A different rotation draw could flip nominal significance, so the result should be read as "the encoded benchmark is detectable", not as a precise probability.

4.1 Partition sensitivity

Block	Blocks n	Full PR-AUC	H2 benchmark
15°	236	0.186	0.068
20°	151	0.139	0.131
30°	72	0.089	0.111
45°	32	0.049	0.085
60°	18	0.084	0.133

The sequence is non-monotone, and post-hoc K-Means folds range from 0.033 to 0.240. It would be unjustified to recast this instability as a stable negative under spatial blocking. The direct finding is that the estimate depends strongly on partition choice while the range/block diagnostic fails.

4.2 Post-hoc USGS mare-domain sensitivity

An external USGS Unified Geologic Map GIS v2 mask uses exact `FIRST_Unit` symbols `Em`, `Im1`, `Im2`, and `Imd`. Folds are assigned before filtering, and the comparison uses raw row-local TiO_2 only; buffers are excluded. In the legacy Imbrian scope, 6,232 pixels, 58 positives, and nine nominal blocks remain. TiO_2+controls PR-AUC is 0.1252 versus 0.0576 for controls, mean increment +0.0676. The five fold increments are 0.2615, 0.0834, −0.0018, −0.0070, and 0.0020; one-sided paired Wilcoxon $p = 0.21875$.

Using the same inherited folds and the same raw-Ti model in both scopes, full-scope controls/controls+TiO_2 R^2 are 0.2590/0.2703 (increment $+0.0113 \pm 0.0049$), while mare-scope values are 0.4257/0.4384 (increment $+0.0127 \pm 0.0126$). The descriptive common-fold increment difference is only +0.0014, and one mare fold is negative. The mare-scope range is about 3,289 km versus 910 km blocks, with effective-region estimate 1.

The mean increment is positive but not statistically significant ($p = 0.219$) and is concentrated in two folds. This post-hoc result is **inconclusive**, not confirmation. The 1:5M mapped-geology mask is also a generalized proxy with source- and rasterization-dependent boundaries.

4.3 Post-hoc H2 injection-recovery power

Thirty phase-randomized nuisance fields per coefficient approximately preserve the spectrum of `log1p`-transformed observed magnetism on the real Imbrian mask and 5 nT prevalence. The encoded H2 antipode-proximity score is added at standardized latent coefficients 0, 0.5, 1, 1.5, 2, 3, and 4. The fixed XGBoost is compared with the same model after H2 ablation. Primary recovery requires a positive mean PR-AUC drop and paired one-sided $p < 0.05$ at 30°; spatially robust recovery also requires a positive drop at 60°.

Latent coefficient	Robust recovery	Wilson 95% CI
0	3/30 (0.1000)	0.0346-0.2562
0.5	18/30 (0.6000)	0.4232-0.7541

Latent coefficient	Robust recovery	Wilson 95% CI
1.0	27/30 (0.9000)	0.7438-0.9654
1.5	28/30 (0.9333)	0.7868-0.9815
2.0	29/30 (0.9667)	0.8333-0.9941
3.0, 4.0	30/30 each	0.8865-1.0000

The point 80% tested-grid minimum detectable coefficient is 1.0 (bracket 0.5-1.0); requiring the Wilson lower bound itself to exceed 80% gives 2.0. This is an extreme floor. At coefficient 1.0, the median positive-rate difference between the top and bottom signal quartiles is 0.05565 and the corrected odds ratio is about 606.8, with effectively no bottom-quartile positives.

A downward grid extension (`h2_antipode_low_strength_extension.json`; strengths 0–0.3, 30 simulations each) maps the low-strength behavior on this axis: the design's zero-signal floor (with-control PR-AUC ≈ 0.135) already exceeds the observed scores (0.089 / 0.111), and across these strengths (injected scores 0.135–0.251) robust detection is only 0.133–0.267 against the 0.100 zero-strength rate. Even the H2 arm therefore has no demonstrated power at realistically-sized effects — its 90%-power operating point corresponds to injected scores several times anything observed. The mean ablation drop is +0.030 **at zero strength** (removing the antipode feature hurts under pure clustered noise), which is the measured mechanism of that arm's anti-conservative false-positive rate, and the mirror image of the H1 arm's −0.011. The matched-null benchmark recovery reported above must be read in this light: a fragile rotation-test result in a score regime where the injection-based criterion fires at most 27% of the time.

The curve is a direct H2-ablation test, not power for the complete publication rule. It does not resimulate nested tuning, permutation, SHAP, or every conjunctive H1 criterion. The coefficient is a simulation unit, not a lunar-physics effect size. The artifact also reproduces a 3,751.7 km fitted range and effective-region estimate of 1.0; both 30° (909.7 km) and 60° (1,819.4 km) blocks remain shorter than the range.

4.4 Completed H1 injection curve and illustrative anchor

The same injection design applied to the **hypothesis actually under test**, with the strength grid extended down into the realistic score regime (`Paper-and-Pitch/h1_tio2_power_analysis.json`, 30 simulations per strength):

Latent strength	Robust recovery	Wilson 95% CI	With-control PR-AUC	Ablation drop
0	1/30 (0.033)	0.006–0.167	0.135	−0.0110
0.025–0.2	0–1/30 each	≤ 0.167	0.131–0.145	−0.003 to −0.012
0.3	0/30	0.000–0.114	0.175	+0.0127
0.5	6/30 (0.200)	0.095–0.373	0.291	+0.0772
1.0	12/30 (0.400)	0.246–0.577	0.622	+0.2795

The tested-grid 80% minimum detectable effect is **not reached at any strength**. The injection design's zero-signal floor (with-control PR-AUC ≈ 0.135) already exceeds the observed scores (0.089 / 0.111); across the low-strength grid (0–0.2, injected scores 0.131–0.145) detection sits at or below the nominal false-positive rate. Mechanistically, the ablation drop is **negative** at weak strengths — matching the observed real-data drop of −0.0030 — because surface-TiO_2 geography is spatially collinear with the `nearside`/`abs_latitude` controls: after ablation, the controls absorb most of a true TiO_2-driven signal. The registered ablation criterion is therefore structurally insensitive to H1 in the weak-signal regime, a design property no seed, fold, or tuning choice repairs.

Zero-strength false-positive rates are **arm-specific**: 0.133 for the H2 arm (the smooth antipode kernel can align with clustered noise blobs spanning CV blocks) versus 0.033 (nominal) for the H1 arm. Neither arm's operating characteristics may be quoted for the other — this also resolves the apparent evidential asymmetry between recovering the H2 benchmark and declining to interpret the H1 null: the H2 arm shows **demonstrated sensitivity** at its operating point, while the H1 arm **did not reach adequate power at any tested strength** (recovery 0.20 at 0.5, 0.40 at 1.0 — inadequate, not zero).

Interpretive anchor (Amendment A8). No physically derived effect size exists — the theory predicts no surface-map effect — so we use latent strength 1.0 as an *illustrative anchor*: the strength at which the same pipeline recovers the H2 geometry control with 90% power. The committed artifact leaves `target_effect_strength` null. Measured H1 power at that anchor is **0.400 (Wilson 95% CI 0.246–0.577)**, far below the 0.80 requirement, and the classification does not depend on it (no tested H1 strength reaches 0.80 power). `adequate_power` is therefore false **on measurement**, not by abstention, and `NOT_SUPPORTED` remains unreachable on evidence.

5 Evidence status and rule history

Status: INCONCLUSIVE_LOW_POWER — now demonstrated, not asserted. The repository-plan success criteria did not pass, so `H1_Supported` is false. The H1 injection curve measures power 0.400 (Wilson 95% CI 0.246–0.577) at the declared strength-1.0 target — a signal seven times stronger than anything observed — and detection at realistic strengths sits at the false-positive rate. Failed H1 criteria are therefore quantitatively uninformative about H1′s truth. The observed H2 encoding is recovered against its matched null, which demonstrates arm-specific sensitivity, not evidence of H1′s absence.

The first adequacy-aware output used `INCONCLUSIVE_SPATIAL_AUTOCORRELATION`. After the result was observed, an asymmetric rule changed a criteria-failing run to `NOT_SUPPORTED` because leakage generally inflates scores. That post-hoc rule did not address low power. The dated amendment now reserves `NOT_SUPPORTED` for a criteria-failing run with demonstrated adequate injection-recovery power. A would-be positive that fails independence retains `INCONCLUSIVE_SPATIAL_AUTOCORRELATION`.

6 Researcher degrees of freedom

The 5/10 nT thresholds, 25/50/100 km buffers, exploratory 600⟶40 km gravity filter, six-basin catalogue, age masks, 30° blocks and block sweep, and model search spaces are reproducible analyst choices, not uniquely forced physical constants. The 300 km antipode length is used by the synthetic generator and H2 injection score, not the observed H2 distance feature. Optuna’s post-hoc maximum over 60 trials is uncalibrated for selection and has no inferential comparison with the repository-plan unselected null.

The configured seed now reaches XGBoost subsampling and rotation sampling. Across ten arbitrary seeds, full PR-AUC is 0.0916 ± 0.0055 and the clean Ti-derived ablation changes sign (mean +0.0032 ± 0.0068). This is a software/RNG diagnostic labelled `NO_INFERENCE`, not independent lunar uncertainty.

7 Follow-up analyses

Injection-recovery power. Both arms are now complete: the direct H2 ablation curve and the H1 (TiO_2-driver) curve with a downward-extended grid. Neither is full pipeline-decision power (tuning, permutation, SHAP, and the conjunctive H1 rule were not resimulated), and the latent coefficient is a simulation unit, not a lunar effect scale. A target effect is now declared (strength 1.0, Amendment A8) with measured H1 power 0.400 — below the 0.80 adequacy requirement, making the low-power classification a measurement.

Terrain validity. The USGS mare-domain comparison above is now available as a post-hoc descriptive sensitivity. Its small positive count and fold concentration do not provide an independent confirmation or repair the structural effective-region limitation.

8 Reproducibility is not inferential adequacy

Acquisition hashes, fail-closed ingestion, package records, deterministic seeds, and tests provide engineering rigor. They do not make pixels independent, validate highland TiO_2, or increase statistical power. The dated amendments are explicitly post hoc and preserve the original text.

This is an **independent learning project**: the author is not a professional planetary scientist and undertook the work because the Nichols et al. (2026) hypothesis was fascinating. Large language models were used extensively — for code, analysis design, documentation, and the adversarial reviews behind the amendments. No AI output is treated as scientific evidence; responsibility for every analysis choice and claim remains with the author, and the complete decision history, including corrected mistakes, is preserved in the repository rather than hidden.

9 Conclusion

The current global rasters and pipeline do not distinguish the tested present-day surface- composition proxy from spatial null structure, and the completed H1 injection curve shows the design has measured, inadequate power (0.400 at the declared target; at the false-positive rate in the realistic regime) to detect even strong versions of that proxy. This is not a discovery of the scale at which titanium "stops" mattering, not a refutation of impact magnetization, and not a test of the temporal intermittent-dynamo mechanism. What the study now contributes is a calibrated statement of what this design **could** and **could not** have seen — the prerequisite any future co-location test must satisfy before its verdict counts.

10 References

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